

Function get bookinfo () {

~~\$htmltable = ''~~

"'search.php <?php \$-post
[id']

\$htmltable = '<tr>';

~~\$htmltable = '<tr> <td> ' . \$booktitle . ' </td> <td> ' . \$bookauthor . ' </td> <td> ' . \$booksubject . ' </td> <td> ' . \$bookquantity . ' </td> <td> ' . \$bookprice . ' </td> </tr>'~~ plug in here

\$linktitle = " " . \$booktitle . ' ';

~~\$htmltable = '<tr> <td> ' . \$linktitle . ' </td> <td> ' . \$bookauthor . ' </td> <td> ' . \$booksubject . ' </td> <td> ' . \$bookquantity . ' </td> <td> ' . \$bookprice . ' </td> </tr>'~~

\$htmltable .= '<td>';

\$htmltable .= \$linktable;

\$htmltable .= '<td>';

\$htmltable .= '<td>'. \$bookauthor . '<td>';

\$htmltable .= '<td>'. \$booksubject . '<td>';

\$htmltable .= '<td>'. \$bookquantity . '<td>';

\$htmltable .= '<td>'. \$bookprice . '<td>';

\$htmltable .= '<tr>';

return \$htmltable;

Question 2 [15 marks]

Create a static HTML page called `book.html`, consisting of a form with an `id` attribute of `bookSearch`, and use the `fieldset` and `legend` tags to group the form's elements. It must have the following input elements, which are all mandatory. *Required*

- Two text inputs allow users to search books by title, author, or both. Name the input elements `booktitle` and `bookauthor`, respectively. The user input must be in alphabetical characters.
- One selection input, its name is `Subjects`, of the book, such as (Computer Science, Mathematics, Physics, Civil Engineering, etc.); one subject must be selected.
- The form uses the `POST` method to send the data to a script titled `search.php` which is hosted at the following domain, `birzeit.edu.ps`, within a folder called `lib`.
- Descriptive labels must be used to make the input elements accessible. You should also use HTML5 form validation attributes to validate user input.

`<!doctype><html>`

`<head>`
`</head>` `<title> book page </title>`

`<body>`

~~`<form action = "birzeit.edu.ps / lib / search.php" method = "post" id = "bookSearch">`~~

`<form action = "birzeit.edu.ps / lib / search.php" id = "bookSearch" method = "post">`

`<fieldset>`

`<legend> books Form </legend>`

`<section>`

`<p>`

~~`<input type = "text" id = "title" name = "booktitle" required>`~~

`<label> title </label>`

`<input for = "title" type = "text" name = "booktitle" required>`

`</p>`

`<p>`

`<label id = "authorname" > author </label>`

`<input for = "authorname" type = "text" name = "bookauthor" required>`

```
$bookress = new book ($var[0], $var[1],  
$var[3], $var[4], $var[5]);
```

??

```
<table>
```

```
<th> book id <th1>
```

```
<th> book title <th1>
```

```
<th> book subject <th1>
```

```
<th> book author <th1>
```

```
<th> quantity <th1>
```

```
<th> price <th1>
```

```
if (!isempty ($book res,
```

```
$bookress → get book info (j)
```

```
} else?
```

```
echo "no record found";
```

~~<table>~~

```
<table />
```

```
<?php
```

```
if (!isset ($-post ['book author']) || !isset ($-post  
['book title'])) { // when empty var.
```

```
header (location: book.html);
```

```
}  
} // close the post method
```

```
$
```

Question Four [25 marks]

Suppose you have a database named library already created and a books table with data already inserted. The database is located on the localhost server and has a username mahe and comp@exam password. The books table has the following columns: id, title, subject, authors, quantity, and price.

Write a PHP script titled search.php that does the following:

- Accepts the data sent by the bookSearch form, as explained in Question 2. The script receives the data via the post method. Then, it connects to the database using PDO and a prepared statement to retrieve the related book details. The data should be fetched to a Book Object, assuming the Book class shown in Figure 1 is given to you in the book.php script, which should be included and used in your script. The search.php script dynamically generates an HTML page that displays the result in a table consisting of title, author, subject, quantity, and price. The book title should be linked to search.php combined with a query string containing a book ID. If no records are found, an HTML page should be displayed to inform the user that no records have been found that match the search criteria.
- If the script (search.php) receives an empty query string, the user should be directed to the search book.html document stored in the same folder as the search.php.
- If the user sends the book ID as a query string using the get method, the script (search.php) generates an HTML page with a single-row table filled with book details.

```
</doctype>
<html>
  <head>
    <title> book search answer </title>
  </head>
```

```
<body>
  <?php
    if (method["method-request"] = post) {
      $book title = $post ['book title'];
      $book author = $post ['book author'];
      if ($isset) {
        $book title = $post ['book title'];
        $book author = $post ['book author'];
      }
    }
  }
}
```

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- If the user sends the book ID as a query string using the get method, the script (search.php) generates an HTML page with a single-row table filled with book details.

```
</doctype>
<html>
  <head>
    <title> book search answer </title>
  </head>
```

```
<body>
  <?php
    if (method ["method-request"] = post) {
      $book title = $post ['book title'];
      $book author = $post ['book author'];
      if (isset ($book title)) {
        $book title = $post ['book title'];
        $book author = $post ['book author'];
      }
    }
  }
  10
```

11. Which of the following PHP scripts correctly retrieves the file type from a form containing a file input field named 'homework'?

- a)

```
<?php
if (isset($_FILES["homework"]) && $_FILES["homework"]["error"] == UPLOAD_ERR_OK) {
    $fileType = $_FILES["homework"]["type"];
    echo "File type: " . $fileType;
} else
    echo "No file uploaded or an error occurred.";
?>
```
- b)

```
<?php
if (isset($_FILES["file"]) && $_FILES["file"]["error"] == UPLOAD_ERR_OK) {
    $fileType = $_FILES["file"]["type"];
    echo "File type: " . $fileType;
} else
    echo "No file uploaded or an error occurred.";
?>
```
- c)

```
<?php
if (isset($_FILES["upload"]) && $_FILES["upload"]["error"] == UPLOAD_ERR_OK) {
    $fileType = $_FILES["upload"]["type"];
    echo "File type: " . $fileType;
} else
    echo "No file uploaded or an error occurred.";
?>
```
- ☒ d)

```
<?php
if (isset($_FILES["homework"])) {
    $fileType = $_FILES["homework"]["type"];
    echo "File type: " . $fileType;
} else
    echo "No file uploaded.";
?>
```

12. One of the following will always add a new element to a PHP array \$a:

- a) `$a[count($a)] = 'New Value';`
- ☒ b) `$a[] = 'New Value';`
- c) `$a = ['New Value'];`
- d) `$a.add('New Value');`

Question 2 [15 marks]

Create a static HTML page called `book.html`, consisting of a form with an `id` attribute of `bookSearch`, and use the `fieldset` and `legend` tags to group the form's elements. It must have the following input elements, which are all mandatory. *Required*

- Two text inputs allow users to search books by title, author, or both. Name the input elements `booktitle` and `bookauthor`, respectively. The user input must be in alphabetical characters.
- One selection input, its name is `Subjects`, of the book, such as (Computer Science, Mathematics, Physics, Civil Engineering, etc.); one subject must be selected.
- The form uses the `POST` method to send the data to a script titled `search.php` which is hosted at the following domain, `birzeit.edu.ps`, within a folder called `lib`.
- Descriptive labels must be used to make the input elements accessible. You should also use HTML5 form validation attributes to validate user input.

```
<!doctype><html>
```

```
<head>  
</head> <title> book page </title>
```

```
<body>
```

```
<form action = "birzeit.edu.ps/lib/search.php" method = "post" action = "birzeit.edu.ps/lib
```

```
<form action = "birzeit.edu.ps/lib/search.php" id = "bookSearch"  
method = "post">
```

```
<fieldset>
```

```
<legend> books Form books Form </legend>
```

```
<section>
```

```
<p>
```

```
<label> title </label>
```

```
id = "title"
```

```
<input for = "title" type = "text" name = "booktitle" required
```

```
</p>
```

```
<p>
```

```
<label id = "authorname"> author </label>
```

```
<input for = "authorname" type = "text" name = "bookauthor"  
required >
```

Question 2 [15 marks]

Create a static HTML page called `book.html`, consisting of a form with an `id` attribute of `bookSearch`, and use the `fieldset` and `legend` tags to group the form's elements. It must have the following input elements, which are all mandatory. *Required*

- Two text inputs allow users to search books by title, author, or both. Name the input elements `booktitle` and `bookauthor`, respectively. The user input must be in alphabetical characters.
- One selection input, its name is `Subjects`, of the book, such as (Computer Science, Mathematics, Physics, Civil Engineering, etc.); one subject must be selected.
- The form uses the `POST` method to send the data to a script titled `search.php` which is hosted at the following domain, `birzeit.edu.ps`, within a folder called `lib`.
- Descriptive labels must be used to make the input elements accessible. You should also use HTML5 form validation attributes to validate user input.

```
<!doctype><html>
```

```
< head>  
< head /> <title> book page < /title />
```

```
< body>
```

```
< form action = "birzeit.edu.ps / lib / search.php" method = "post" action = "birzeit.edu.ps / lib
```

```
< form action = "birzeit.edu.ps / lib / search.php" id = "bookSearch"  
method = "post" >
```

```
<fieldset>
```

```
< legend> books Form < legend />
```

```
<section>
```

```
<p>
```

```
< label> title < label />
```

```
id = "title"
```

```
< input for = "title" type = "text" name = "booktitle" required
```

```
< /p>
```

```
< p>
```

```
< label id = "authorname" > author < label />
```

```
< input for = "authorname" type = "text" name = "bookauthor"  
required >
```

\$author, quantity, price) {

3

~~id, title, subject~~

'id' → this.\$BookID;

'title' → this.\$book title;

'subject' → this.\$bookSubject;

'author' → this.\$Book-author;

'quantity' → this.\$quantity;

'price' → this.\$price;

}

// generate setters + getters example I'd do one but all are
the same Sorry I didn't have enough
time

behind

Function getURL () {

2

\$link = '<a';

\$link .= " href = 'search.php?";

> " > ~~link for~~ name = "book title";

> "book id / title link" . "<a />" ;

return \$link;

}

```

<select name = 'Subjects' >
  <option name = 'sub1' value = 'computer science' >
    computer science </option>
  <option name = 'sub2' value = 'mathematics' > mathematics </option>
  <option name = 'sub3' value = 'physics' > physics </option>
  <option name = 'sub4' value = 'civil engineer' > civil eng. </option>
  ...
  <option name = 'sub5' value = 'biology' selected > bio log </option>
</select>
</Section>
  <input type = 'submit' name = 'submit' > submit </input>
</fieldset>
</form>
</body>
</html>

```

// no php code cuz file name extension is
 • .html not • .php.

```

<select name = 'Subjects' >
  <option name = 'sub1' value = 'computer science' >
    computer science </option>
  <option name = 'sub2' value = 'mathematics' > mathematics </option>
  <option name = 'sub3' value = 'physics' > physics </option>
  <option name = 'sub4' value = 'civil engineer' > civil eng. </option>
  ...
  <option name = 'sub5' value = 'biology' selected > bio log </option>
</select>
</Section>
  <input type = 'submit' name = 'submit' > submit </input>
</fieldset>
</form>
</body>
</html>

```

// no php code cuz file name extension is
 • .html not • .php.

Question Three [15 marks]

Write a PHP script called **book.php** to define the Book class, as shown in Figure 1 below.

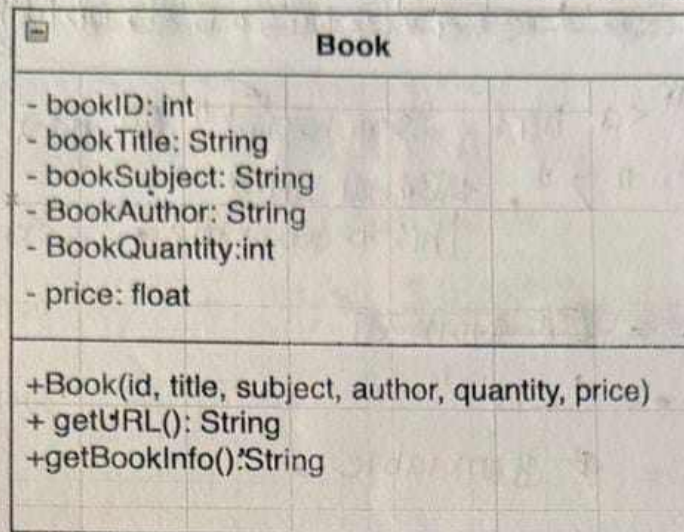


Figure 1: Book Class

- Write a **getURL** method that **returns** (not echos) an HTML link element. The link should point to the *search.php* script combined with a query string containing a book ID and label the link with the book title.
- Write **getBookInfo** method that **returns** the following book details as an HTML table row (<tr>) element, title, author, subject, quantity, and price. The book title should be linked to *search.php* combined with a query string containing a book ID.

<?php

```
private $bookID;  
private $bookTitle;  
private $bookSubject;  
private book $BookAuthor;  
private $BookQuantity;  
private $price;
```

3

13. PDOStatement::fetch function:

- a) Get a row from the result set returned by the PDOStatement
- b) Prepares and executes an SQL statement
- c) Creates a PDO instance to represent a connection to the requested database
- d) Returns the number of rows affected by the last SQL statement

14. Which of the following is the correct SQL statement that is used in a prepared statement and uses only named parameters for all of its substituted (variable) values:

- a) insert into students (name, city) values (\$name, \$city)
- b) insert into students (name, city) values (:name, :city)
- c) insert into students (name, city) values (?, ?)
- d) insert into students (name, city) values (\$_POST['name'], \$_POST['city'])

15. You have given the following HTML form

```
<form method="post" action="process.php">


```

Assume the user selects the first and last option. What is the output of the following PHP script

```
<?php
$a = $_POST['op'];
$output = "";
foreach($a as $v)
    $output .= $v . " ";
echo $output;
```

Handwritten notes: 10 → greek paper, 5 → Red onion

Handwritten output: 10 " 5

- a) 15
- b) Green Paper Red Onion
- c) 10 5
- d) Mushroom

Please copy your answer to the table provided below.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| b | a | c | a | c | c | b | a | c | c | d | b | a | b | c |

33

```
if (isset($_POST['bookauthor']) ||  
    isset($_POST['booktitle'])) {
```

```
    $title = $_POST['booktitle'];  
    $author = $_POST['bookauthor'];
```

```
    try {
```

```
        $connString = "mysql: host = local host, dbname =  
            library";
```

```
        $username = "maher";
```

```
        $password = "comp@exam";
```

```
2 $pdo = new PDO($connString, $username, $pas  
$pdo->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
```

```
$sql = "select * from books where 'booktitle' = "  
    $title . " ";
```

```
$pdo->prepare($sql);
```

```
include "book.php";
```

```
$pdo->prepare($sql);
```

```
$res = $pdo->query($sql);
```

```
$res = $pdo->prepare($sql);  
for ($res as $a) {
```

```
    $b = $a['booktitle'];
```

5. Which of the following options correctly identifies the variables and their values passed to the server in the URL "http://www.abc.com/view.php?action=new&id=50"?
- a) action, view.php, new, id, 50
b) http://www.abc.com
☒ c) action=new, id=50
d) view.php, action, id
6. Which statement correctly retrieves a field named "category" from a form that uses the post method?
- a) \$_POST[category]
b) \$_GET["category"]
☒ c) \$_POST["category"]
d) \$_REQUEST[category]
7. What is the correct way to include the file "time.inc" ?
- a) <?php include:"time.inc"; ?>
☒ b) <?php include "time.inc"; ?>
c) <?php include file="time.inc"; ?>
d) <!-- include file="time.inc" -->
8. Which PHP function returns true if the passed PHP variable exists?
- ☒ a) isset(\$var)
b) exists(\$var)
c) empty(\$var)
d) isempty(\$var)
9. 'PHP_SELF' is:
- a) A string that stores the filename of the currently executing script.
b) A variable that stores the filename of the currently executing script.
c) A superglobal variable that stores the filename of the currently executing script.
d) A key in \$_SERVER that stores the filename of the currently executing script.
10. One of the following deletes only one element from the array \$a = [1/2=>3/4=>'b']:
- a) unset(\$a[1])
b) unset(\$a)
☒ c) unset(\$a[4])
d) unset(\$a[3])

[2] (a)

The 3 key points that characterize the imperative (procedural) paradigm are:

1. Use Variables to describe a memory location.
2. Use statements to change the value of the variable.
3. ~~derive~~ One way to execution statements (no recursive) (sequence, instruction execution).

While the 3 key points that characterize the Object Oriented paradigm are:

1. ~~Inheritance~~ Inheritance: share data and methods to sub-classes.
2. encapsulation: hide the data.
3. Polymorphism: many forms for same instance.

(b) Fill in the blanks:

1. Using names for data types such as "int" and "float" is called Variables.

2. Describing the different structures of the PL is called Syntax, while describing the meaning of those structures is called Semantic.

3. The expression $20-7*3*2$ is written in Lisp as $(- 20 7 (* 3 2))$.

4. In block structured languages the runtime environment technique used is stack based or static.

5. A language with few numbers of instruction is said to be more simplicity.

~~$20-7*3*2$~~ $(- 20 7 (* 3 2))$ $7-20$

7
20
+

~~$20-7*3*2$~~ $(- 20 7) (* 3 2)$